



TRAINING WORKBOOK MODULE 3

A STEP BY STEP GUIDE

ABSTRACT

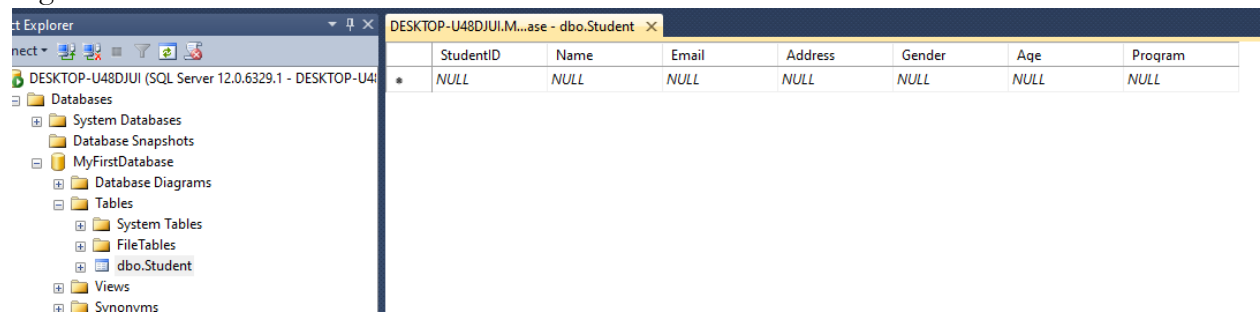
This module the steps required to create tables, indexes and more

NSTAR Technologies

Certificate Program – Database Administration

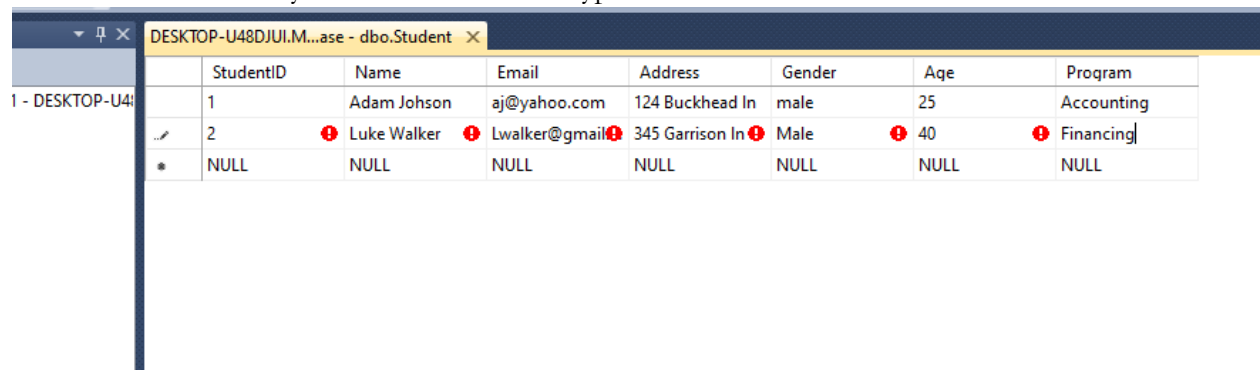
1. Adding records to the table
GUI

- Right click on the table and Select EDIT TOP 200 ROWS



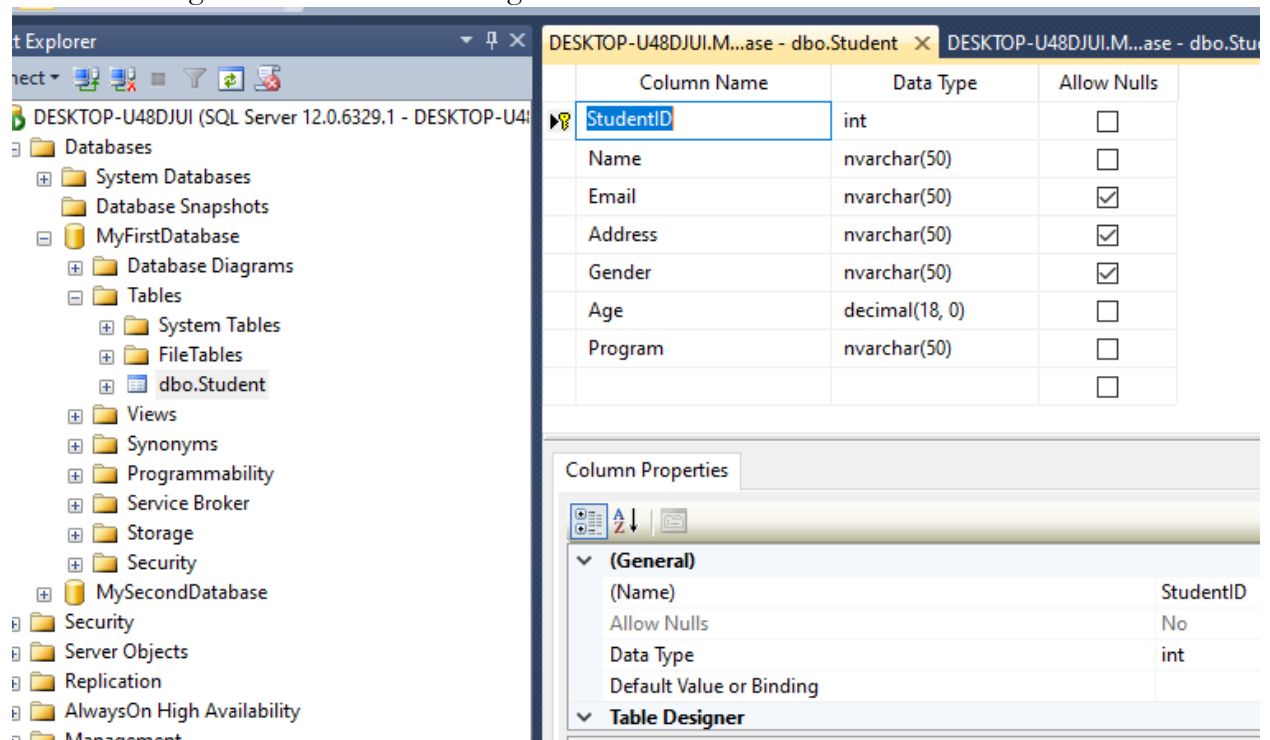
	StudentID	Name	Email	Address	Gender	Age	Program
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- To add values manually click on column and type in value



	StudentID	Name	Email	Address	Gender	Age	Program
1 - DESKTOP-U48DJUI	1	Adam Johson	aj@yahoo.com	124 Buckhead In	male	25	Accounting
	2	Luke Walker	Lwalker@gmail	345 Garrison In	Male	40	Financing
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

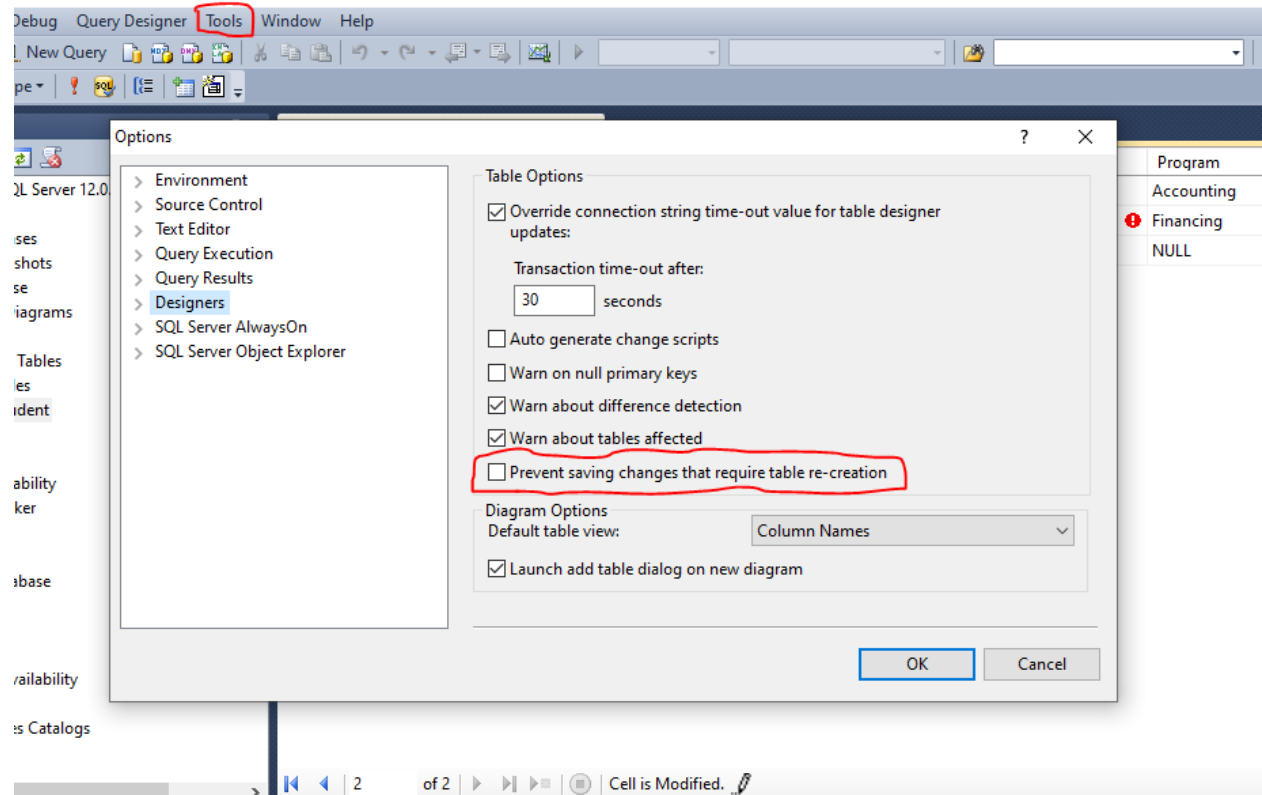
- To make changes to structure of table right click on table and select



Column Name	Data Type	Allow Nulls
StudentID	int	☐
Name	nvarchar(50)	☐
Email	nvarchar(50)	☒
Address	nvarchar(50)	☒
Gender	nvarchar(50)	☒
Age	decimal(18, 0)	☐
Program	nvarchar(50)	☐
		☐

Column Properties	
(General)	
(Name)	StudentID
Allow Nulls	No
Data Type	int
Default Value or Binding	
Table Designer	

- To allow future change to table structure, click on tools select options, click designers, and uncheck “Prevent saving changes that require table re-creation”



2. Downloading random data

- Type/copy and paste <https://mockaroo.com/> into browser

- Click on each field name to rename it


mockaroo realistic data generator

Need some mock data to test your app? Mockaroo lets you generate up to 1,000 rows of realistic test data in CSV, JSON, SQL, and Excel. Download data using your browser or sign in and create your own [Mock APIs](#).
 Need more data? Plans start at just \$50/year. Mockaroo is also available as a [docker image](#) that you can deploy in your own cloud.

Field Name	Type	Options
Student_id	Row Number	blank: 0 % <i>fx</i> ×
first_name	First Name	blank: 0 % <i>fx</i> ×
last_name	Last Name	blank: 0 % <i>fx</i> ×
email	Email Address	blank: 0 % <i>fx</i> ×
gender	Gender	blank: 0 % <i>fx</i> ×
Age	IP Address v4	blank: 0 % <i>fx</i> ×

Add another field

- Click folder looking icon to change contents of Field name

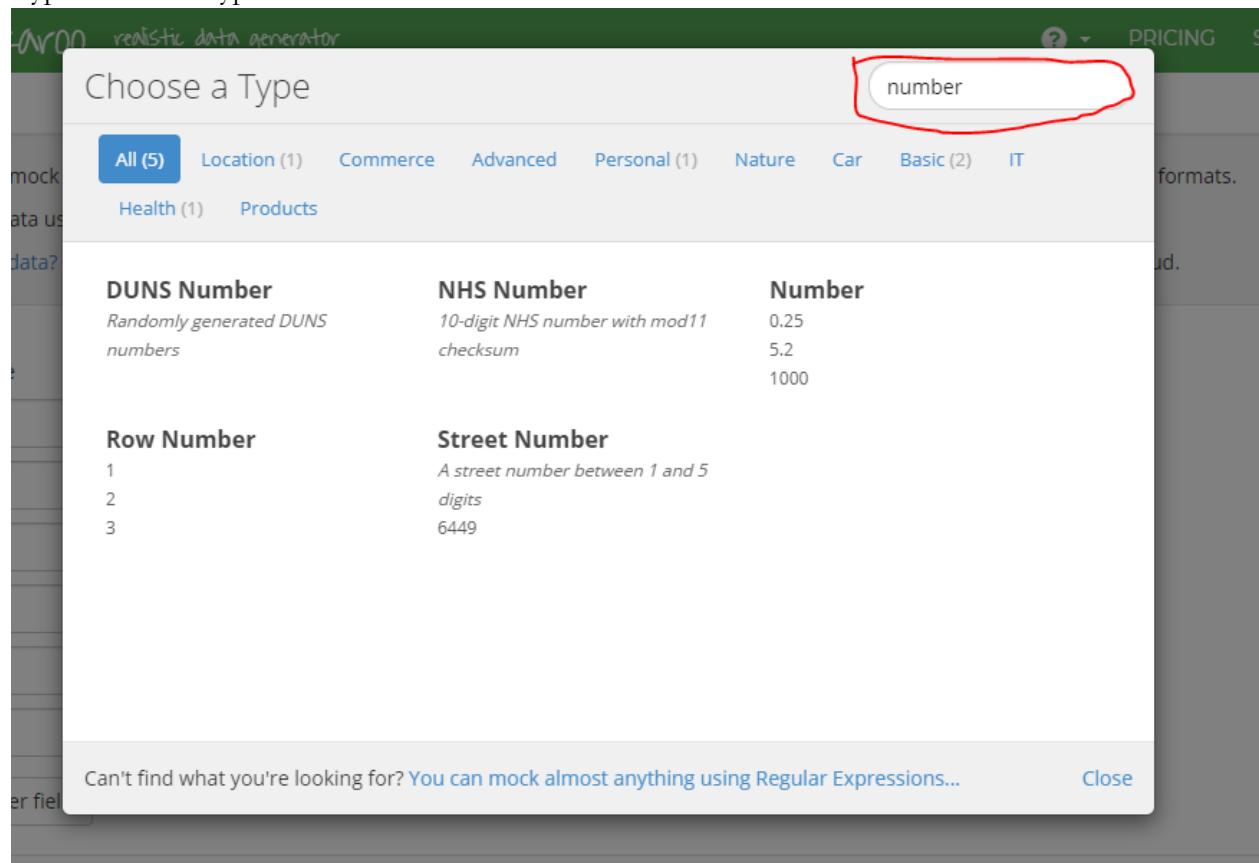

mockaroo realistic data generator

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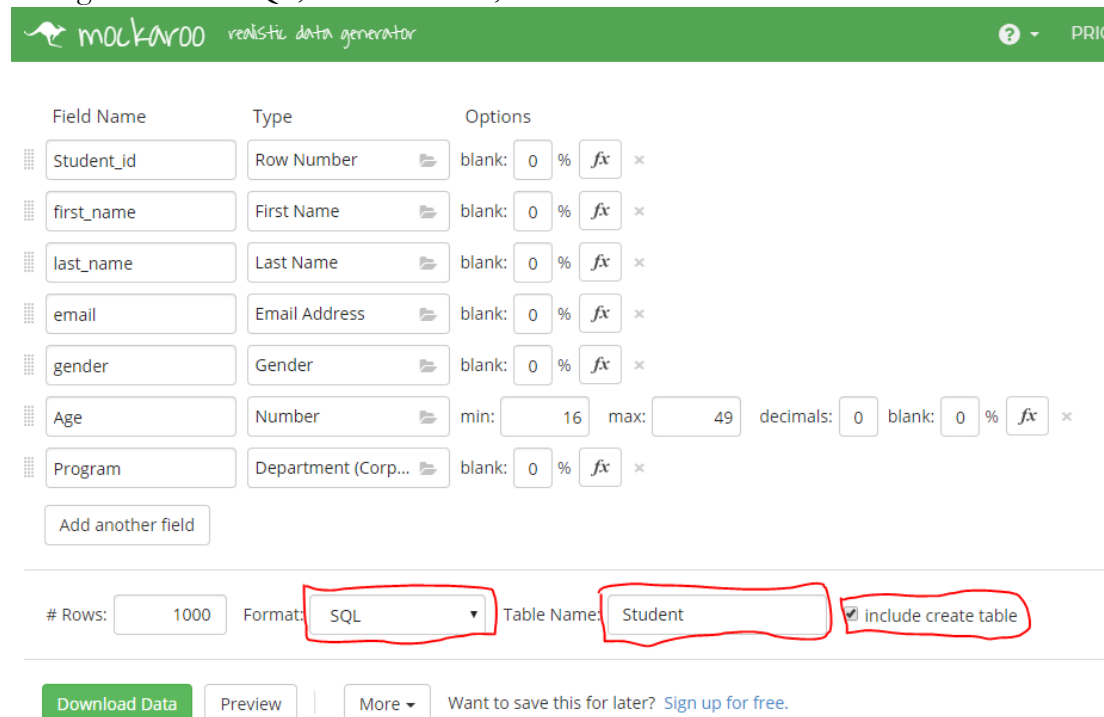
Field Name	Type	Options
Student_id	Row Number	blank: 0 % <i>fx</i> ×
first_name	First Name	blank: 0 % <i>fx</i> ×
last_name	Last Name	blank: 0 % <i>fx</i> ×
email	Email Address	blank: 0 % <i>fx</i> ×
gender	Gender	blank: 0 % <i>fx</i> ×
Age	IP Address v4	blank: 0 % <i>fx</i> ×

Add another field

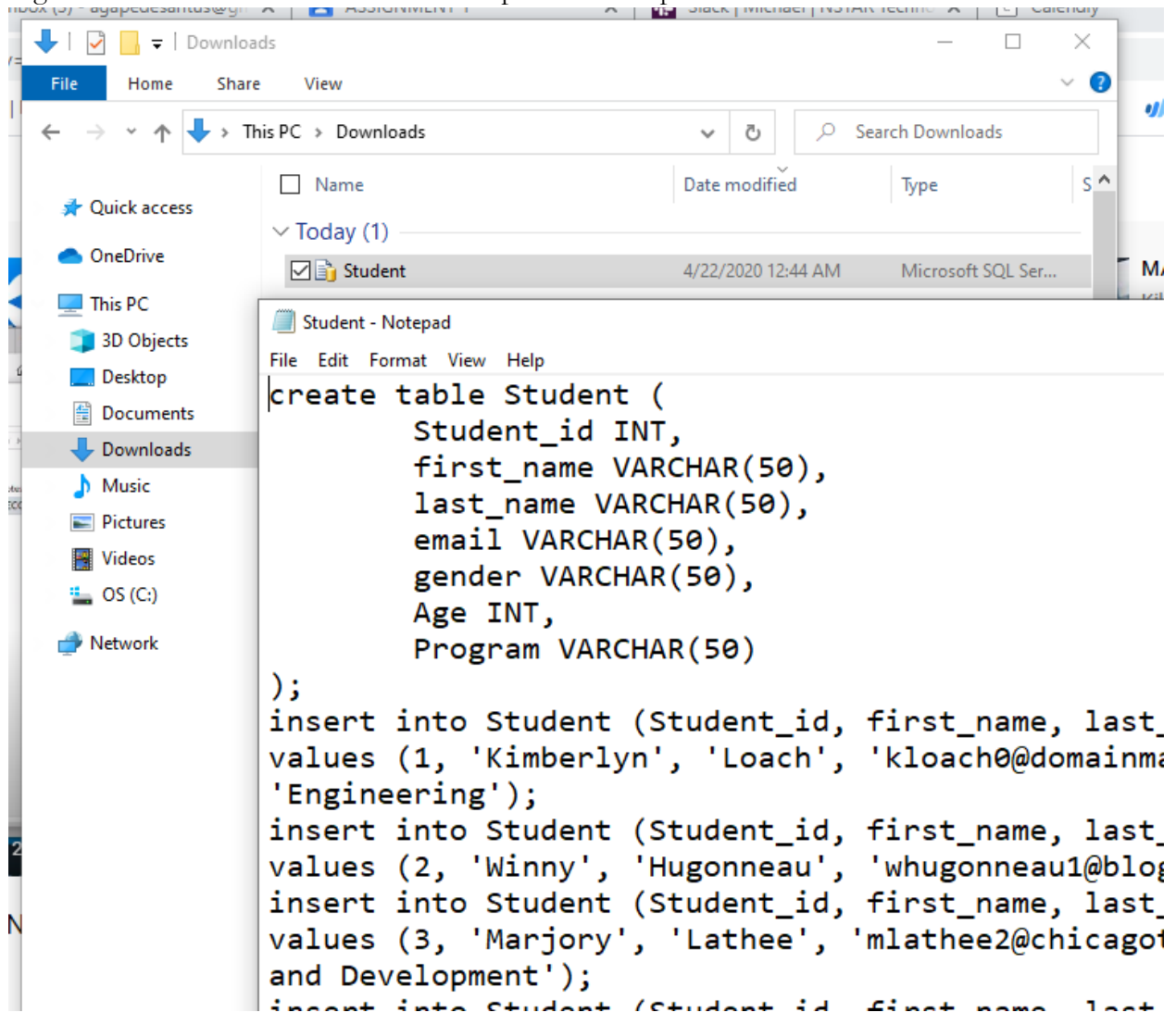
- Type in desired type into search box



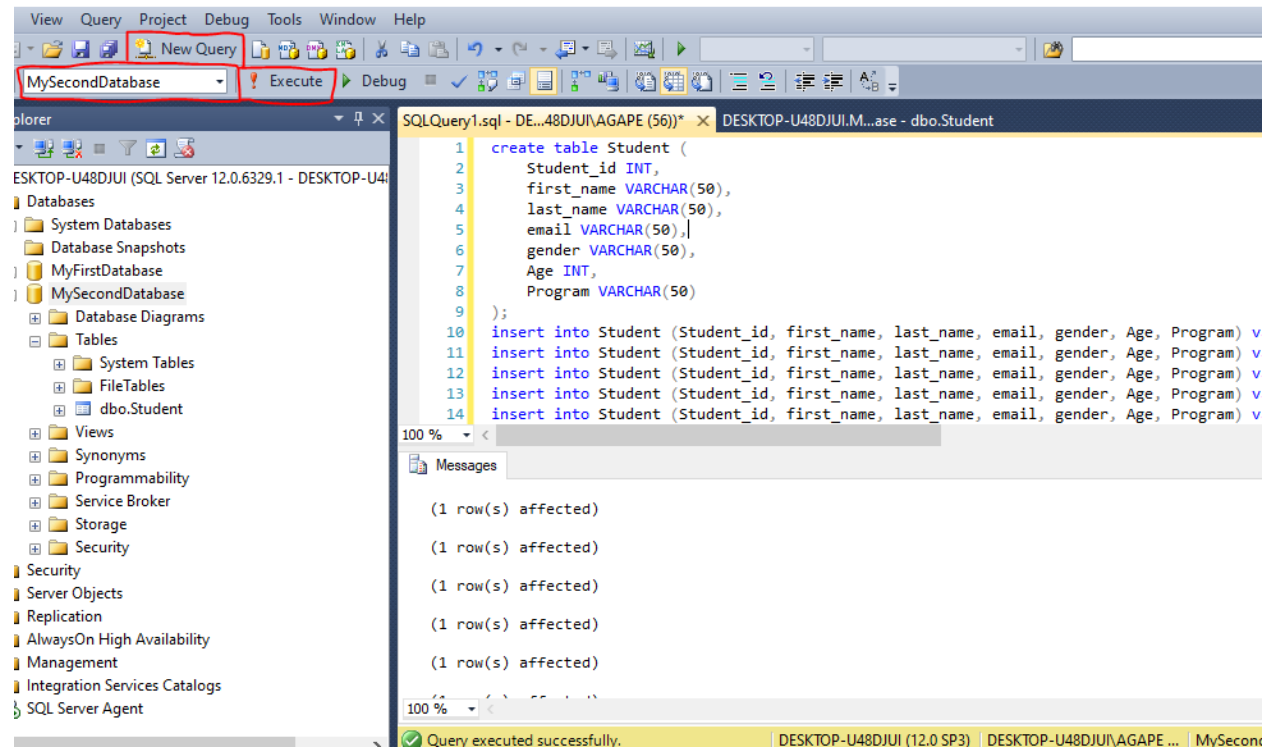
- Change format to SQL, name the table, and check include create table



- Next click download data
- Right click on downloaded file and select open with notepad



- Copy the contents and paste in management studio
- To paste in management studio, Right click on desired database and select NEW QUERY, paste in TSQL script
- Or click on New Query paste in TSQL Script and Specify Database to store table, by changing from MASTER to MyFirstDatabase/MySecondDatabase
- Then click on EXECUTE/F5



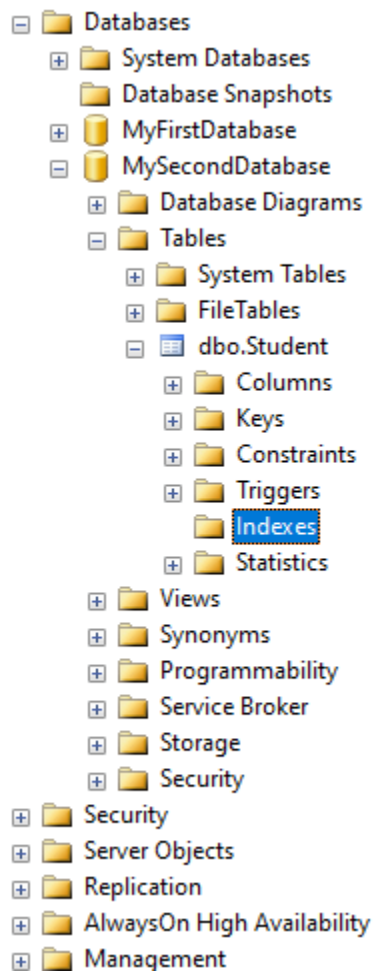
- For confirmation right click on table and select EDIT TOP 200 ROWS

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane shows the database structure for 'DESKTOP-U48DJUI (SQL Server 12.0.6329.1 - DESKTOP-U48DJUI)'. The 'Databases' folder is expanded, showing 'System Databases', 'Database Snapshots', 'MyFirstDatabase', and 'MySecondDatabase'. Under 'MySecondDatabase', the 'Tables' folder is expanded, and the 'dbo.Student' table is selected.

The main pane shows the 'dbo.Student' table with 200 rows. The columns are: Student_id, first_name, last_name, email, gender, Age, and Program. The rows are numbered 1 through 200. The status bar at the bottom indicates '1 of 200' rows are displayed.

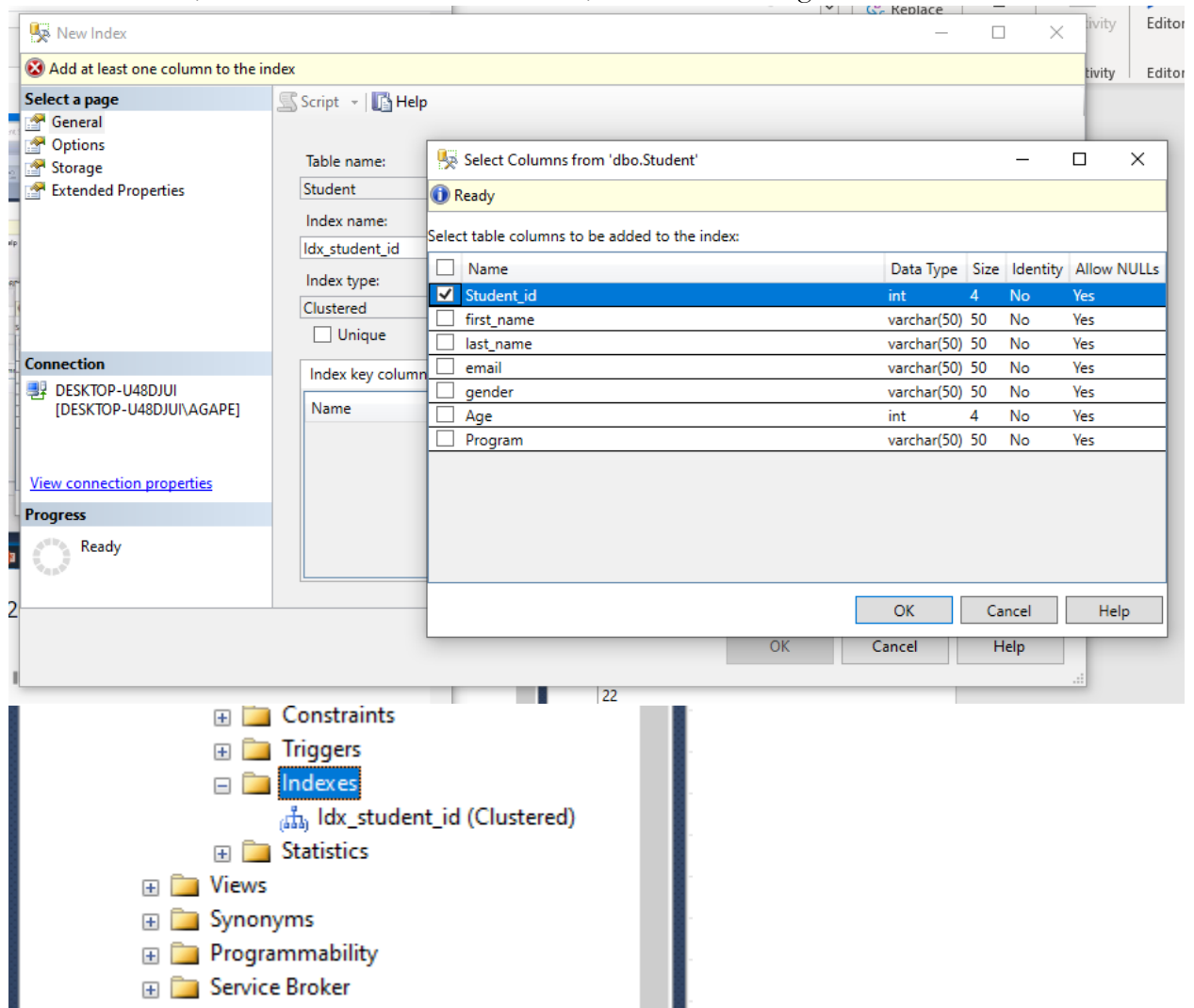
Student_id	first_name	last_name	email	gender	Age	Program
1	Kimberlyn	Loach	kloach0@doma...	Female	25	Engineering
2	Winny	Hugonneau	whugonneau1...	Female	48	Legal
3	Marjory	Lathee	mlathee2@chic...	Female	28	Research and D...
4	Charyl	Redd	credd3@bizjour...	Female	23	Sales
5	Lazaro	Bowmaker	lbowmaker4@n...	Male	29	Services
6	Morrie	Ormes	mormes5@ind...	Male	45	Sales
7	Florida	Heugh	fheugh6@studi...	Female	47	Product Manag...
8	Meredith	Reece	mreece7@gizm...	Male	39	Research and D...
9	Doug	Dunston	ddunston8@ch...	Male	33	Engineering
10	Poul	Drabble	pdrabble9@usn...	Male	31	Support
11	Jerrie	Piecha	jpiecha@nyti...	Female	38	Marketing
12	Quentin	Horning	qhorningb@twi...	Female	26	Accounting
13	Ronni	Matteoni	rmatteonic@ti...	Female	37	Engineering
14	Jerrie	Cracie	jcracied@vk.com	Female	30	Services
15	Leda	Howroyd	lhowroyde@i2i.jp	Female	22	Product Manag...
16	Bart	Doret	bdoret@opera....	Male	18	Product Manag...
17	Nannie	Nealy	nnealyg@sina.c...	Female	44	Services
18	Ashely	Benning	abenningh@his...	Female	35	Support
19	Brewster	Matelyunas	bmatelyunasi@...	Male	33	Training
20	Garnet	Forster	qforsterj@ocn....	Female	43	Human Resour...

3. To create a clustered index, expand tables, expand table created, right click on index, select new index, and select clustered index

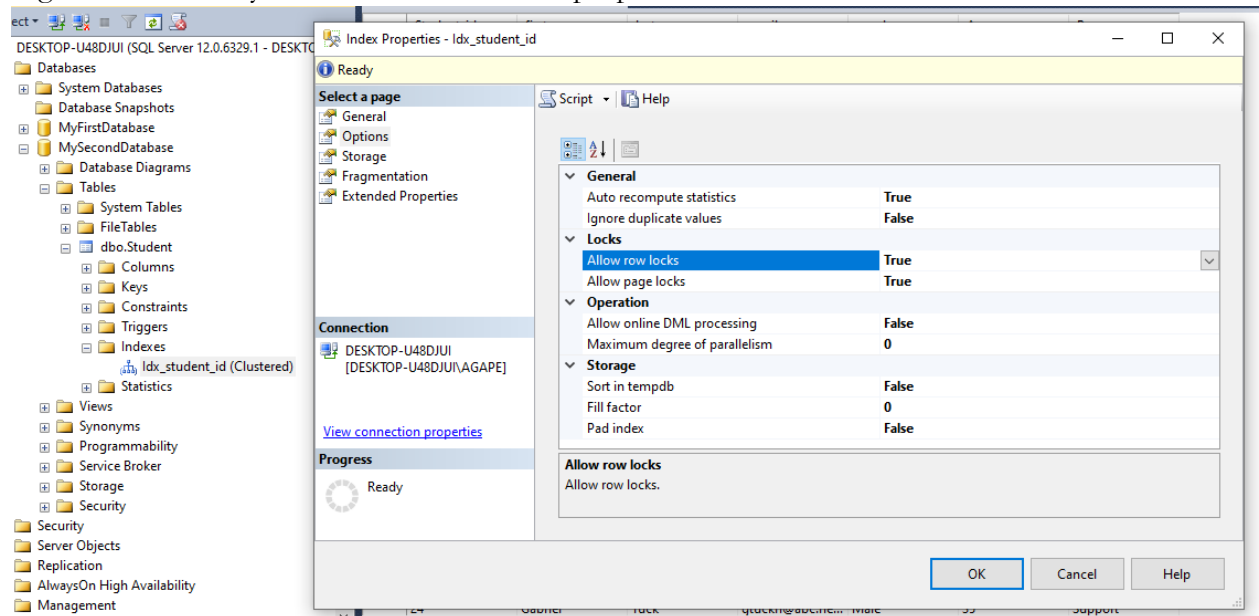


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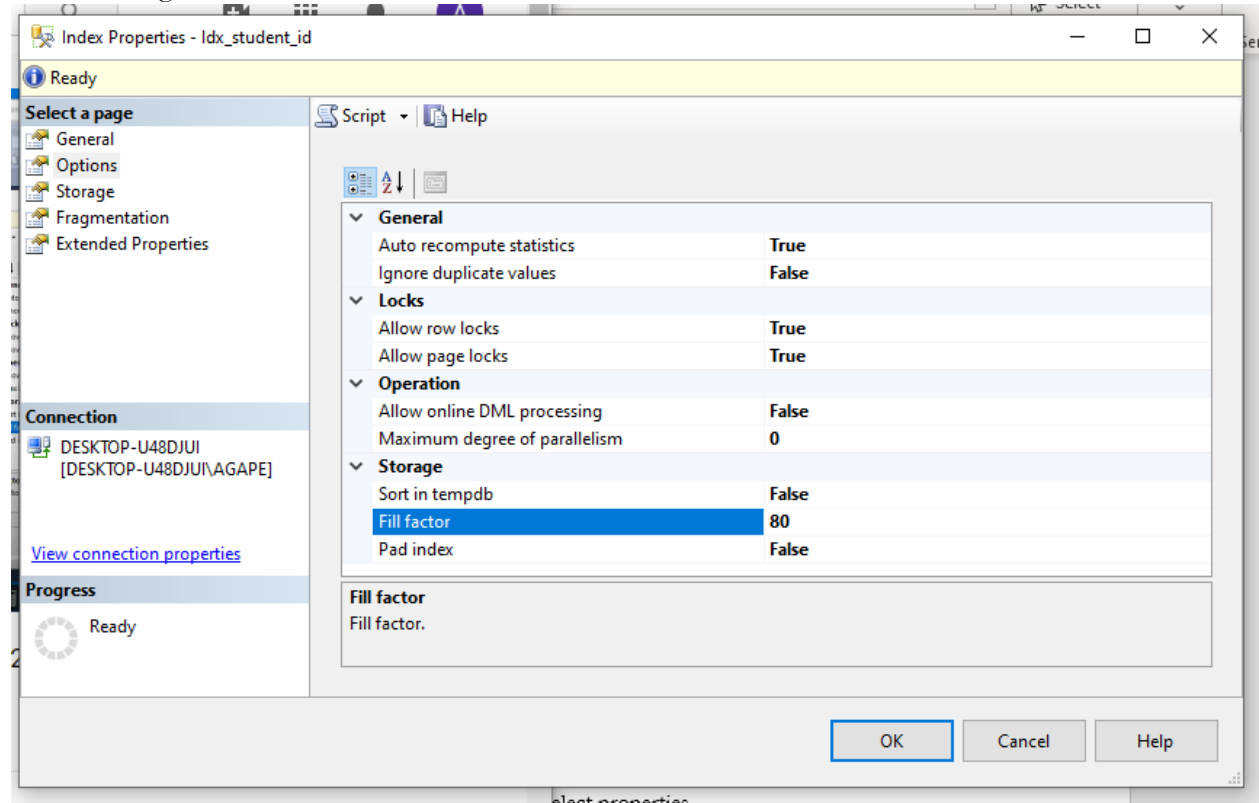
- Name the index, and select column to be indexed, click ok and ok again



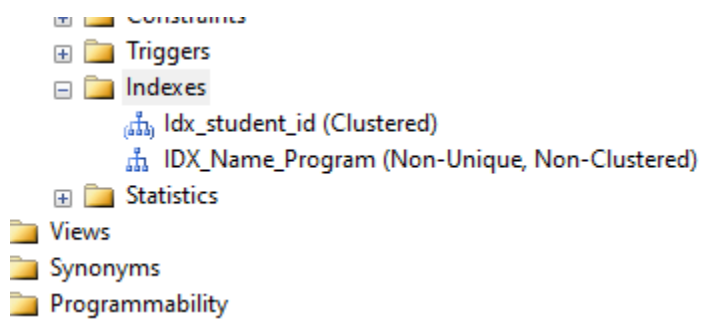
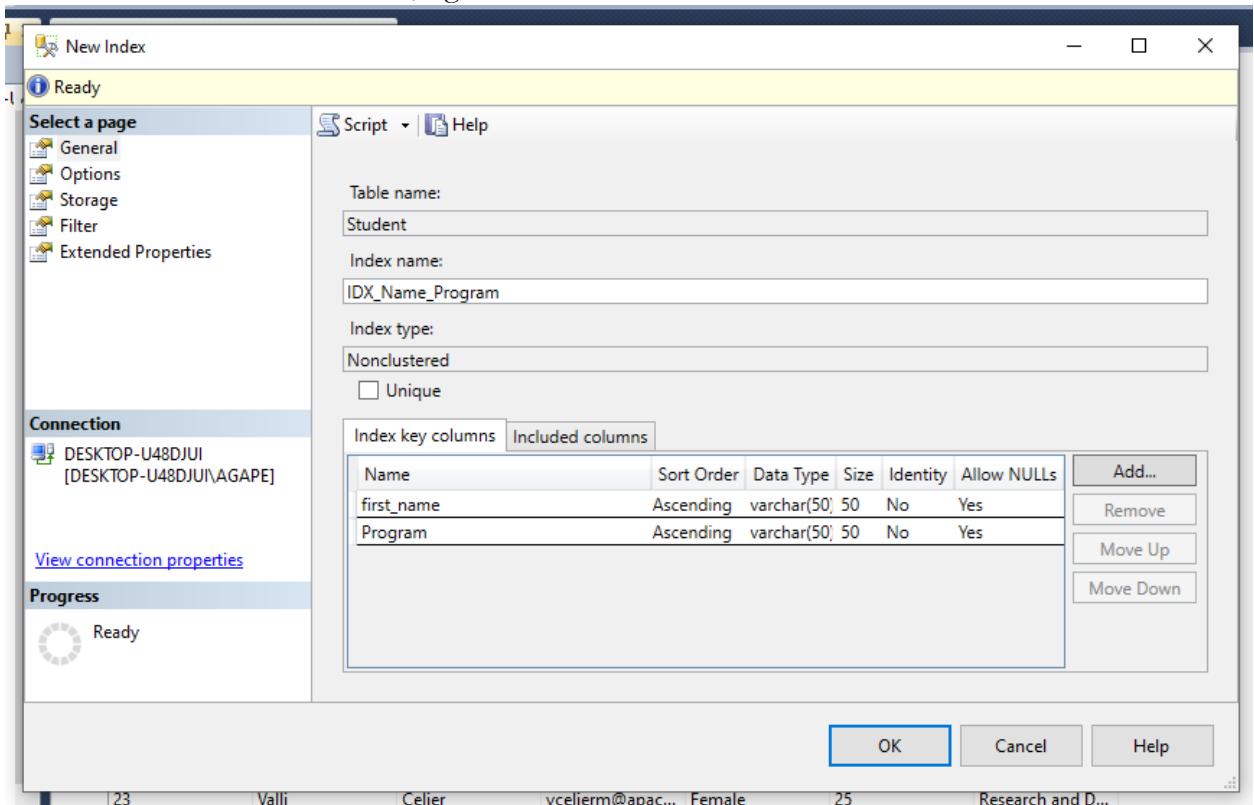
- Right click on newly created index and select properties



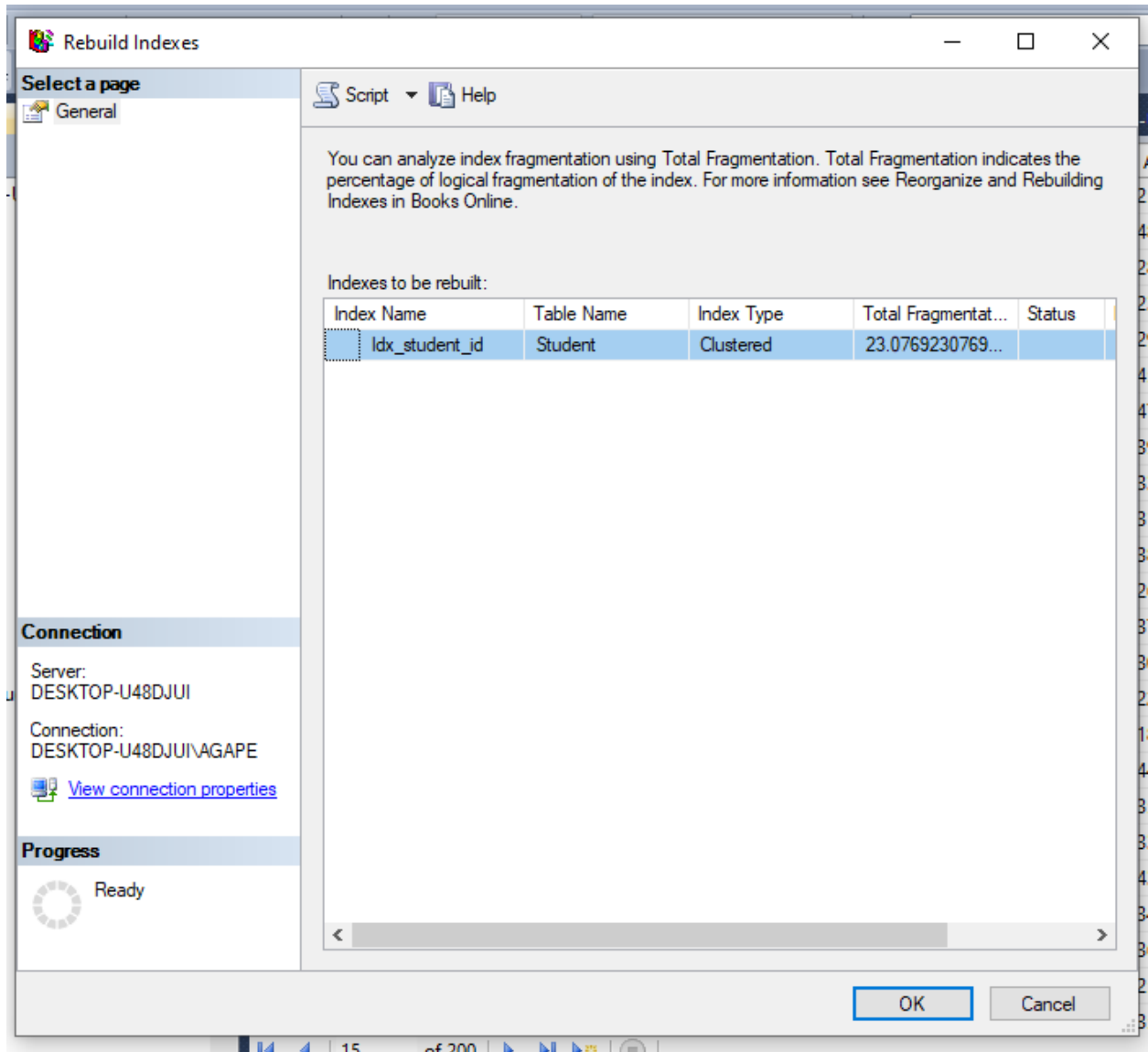
- Under storage set Fill Factor to 80



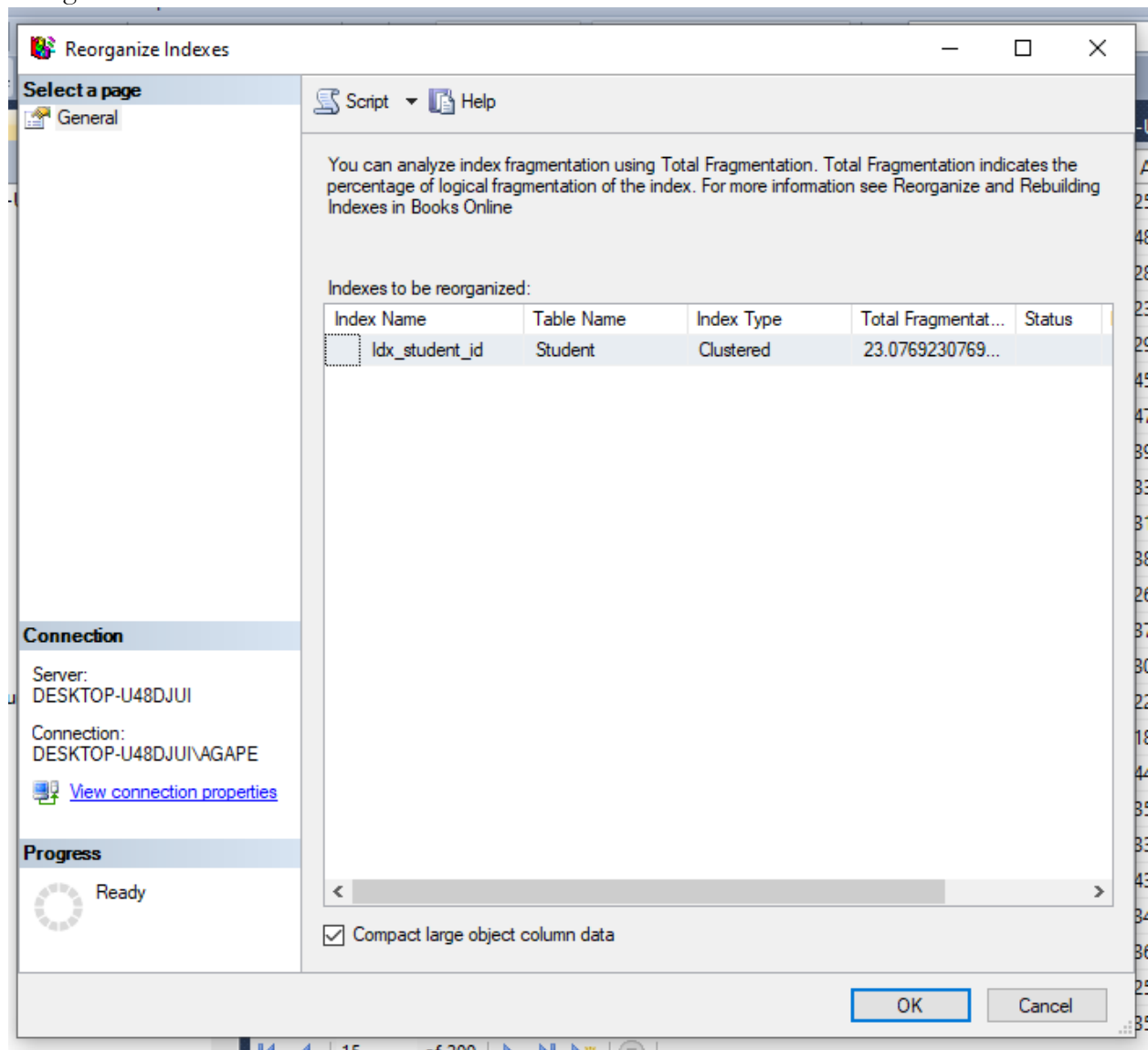
4. To create a Non-Clustered index, right click on index and select Non-Clustered



- To Rebuild or Reorganize and index Right click on index
- Rebuild

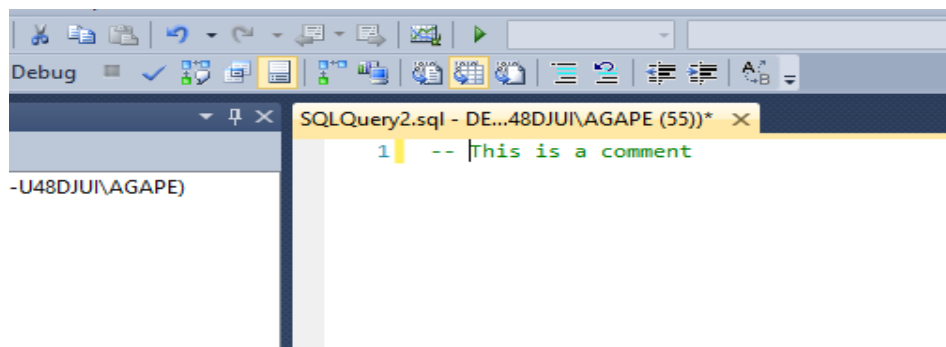


- Reorganize



5. T-SQL SCRIPTS

- Use two dashes to make a comment '--'



- THE T-SQL SCRIPTS WILL BE PROVIDED